BCH 4312 INDUSTRIAL BIOCHEMISTRY

- BIOPROCESS TECHNOLOGY
- ENZYME TECHNOLOGY
- USE OF MICROORGANISMS AS SOURCES OF ENZYMES
- USE OF IMMOBILIZED ENZYMES IN INDUSTRIAL PROCESS
- USE OF GENETIC ENGINEERING IN ENZYME TECHNOLOGY
- ADDITIVES USED IN INDUSTRIAL PROCESSING OF FOODS

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Industrial Biochemistry is the application of chemistry of life in the industry to produce products, mostly in food, pharmaceutical and chemical industries. It is similar to Biotechnology.

FERMENTATION (BIOPROCESS) TECHNOLOGY

The early fermentation (also called bioprocess) technology involves the use of microorganisms for the production of cheeses, yoghurts, fermented pickles and sausages, and beverages such as beers, wines and spirits.

Bioprocess Technology involves multitude of complex enzyme-catalysed reactions within specific microorganisms leading to the production of various products.

Fermented foods are foods that have been intentionally altered during production by introducing certain quantity of organisms (starter culture) and carefully controlling their activity through nutrient availability and modified environmental conditions.

Organisms responsible for fermentations

The most common groups of microorganisms involved in food fermentation are:

1) Bacteria 2) Yeasts 3) Moulds

1) Fermentation by Bacteria

Fermenting bacteria can be grouped into 3:

a. Lactic Acid Bacteria

These are bacteria which convert carbohydrates to lactic acid. They are Gram – positive, non – sporing, obligate fermenters belonging to the following genera: *Lactobacillus, Lactococcus, Streptococcus, Leuconostoc* and *Pediococcus.*

Lactic acid bacteria carry out their reactions – the conversion of carbohydrate to lactic acid plus carbon dioxide and other organic acids, in the absence of oxygen. The species commonly used in lactic acid fermentation are: *Lactobacillus acidophilus, L. bulgaricus, L. caret, L. brevis, L. thermophilus, L. mesenteroides, L. plantarum, L. bovis, L. casei, L. delbrueckii.*

Typical products of lactic acid fermentation include Sauerkraunt (pickled "salted" vegetable), brine salted vegetables, Grunduk (pickled leafy vegetable, fermented tea leaves, yoghurt, cheese, sour milk, fermented cassava, fermented legumes, fermented meat and fermented cereal products. Traditionally fermented milk products include *Sethemi* (South Africa), *Kulenaoto* (Masai – Kenya).

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b. Acetic Acid Bacteria

These are important bacteria in the fermentation of fruits and vegetables. They are Gram-negative, aerobic bacteria belonging Acetobacteraceae family. They oxidize sugars or ethanol to acetic acid. The most important members of this group are *Acetobacter* species (eg *A. aceti, A. cerevisiae, A. malorum, A. oeni A. pasteurianus* etc), *Gluconacetobacter* species (eg. *G. entanii, G. liquefaciens* etc), *Gluconobacter oxydans* and *Komagataeibacter europaeus*. In addition to acetic acid (vinegar) production, they are also used industrially in the production of L-sorbose from D-sorbitol, dihydroxyacetone from glycerol and bacterial cellulose.

c) Alkaline Fermenters

These bacteria ferment protein – rich foods such as soyabeans and other legumes. They belong to the *Bacillus* species, especially *Bacillus subtilis, B. lichenformis* and *B. pumilus*. They hydrolyze the protein in the legumes to amino acids and peptides releasing ammonia, which increases the alkalinity and makes the substrate unsuitable for growth of spoilage organisms. Example is the production of *Iru* and *Daddawa* (traditional food condiments from Nigeria).

2) Fermentation by Yeasts

The most beneficial yeasts in terms of desirable food fermentation are from the *Saccharomyces* family. In the food industry, yeasts produce enzymes that catalyze certain chemical changes such as the leavening of bread and the production of alcohol.
Products of fermentation by yeasts include bread, alcoholic beverages [beers (cereals and fruit beers) and wines (fruit wine, cashew wine, grape wine, date wine, palm wine, sugar cane wine)], ethanol etc.

3) Fermentation by Moulds

Moulds also play a significant role in the desirable fermentation of fruit and vegetable products, eg *Penicillium* (Cheese), *Aspergillus oryzae, A. sojae* (Soy sauce and Rice) and *Rhizopus* (Tempeh)

Products of moulds fermentation include *Koji* (soy sauce), *Tempeh* (soy bean fermentation in Indonesia), *Miso* (rice fermentation in Asia).

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ome food products produced by Fermentation

1) Yoghurt – *Starting Material* – Pasteurized Milk. **Fermenting organism**: Lactic acid Bacteria (eg. *Streptococcus thermophilus* and *Lactobacillus delbrueckii* subspecies *bulgaricus).*

2) Fermented Locust Bean Seed (*Daddawa).* *Starting Material – Parkia biglobisa* seeds . Fermenting organism- *Bacillus* species (*Bacillus subtilis, B. pumilis* and *B. licheniformis)*

3). Bread. Starting Materials: Wheat flour, milk, oil, salt and sugar). **Fermenting Organism**: Yeast (*Saccharomyces cerevisia)*

4). Soy Sauce. Starting Materials: Cooked soy beans and roasted cracked wheat. **Fermenting Organism**: Mould (*Aspergillus oryzae* or *A. sojae).*

OTHER PRODUCTS OF FERMENTATION

In addition to foods and beverages, microbial fermentation is used to produce:

i. Essential primary metabolites, such as acetic acid, lactic acid, organic acids, amino acids, vitamins and polysaccharides.

ii. Secondary metabolites, such as penicillin, streptomycin, gibberellins etc

iii. Industrially Useful Enzymes eg Exocellular Enzymes such as amylases, pectinases and proteases, an Intrarcellular Enzymes such as invertase, asparaginase, restriction endonucleases etc.

Some fermentation products in the food industry include:

- Dairy products (cheese, yoghurt, fish and meat products)
- Beverages (alcoholic, tea and coffee)
- Baker's yeast
- Food additives (antioxidants, colours, flavours, stabilizers)
- Novel foods (soy sauce, tempeh, miso)
- Mushroom products
- Starch products
- Glucose and High-fructose syrups
- Food enzymes

Other products in the Agricultural sector include:

- Animal feedstuffs (single cell proteins)
- Veterinary vaccines

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- Microbial pesticides
- Rhizobium and other nitrogen-fixing bacterial inoculants
- Plant cell and tissue culture (for vegetative propagation, embryo production and genetic improvements).
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Plant Cell and Mammalian Cell Culture

Bioprocess technology is currently using cells derived from higher plants and animals to produce many important products. Plant Cell Culture is aimed largely at secondary products formation, such as flavours and antibiotics; whereas Mammalian Cell Culture has been concerned with vaccine formation and production of protein molecules.

Product Formation Stages in Bioprocess Technology

- The product formation stages in bioprocess technology are essentially similar no matter the organism selected, medium used and product formed.
- Large number of cells is grown under defined conditions, correct medium composition
- The organism must be cultivated and motivated to form the desired products by means of physical technical containment system (the bioreactor)
- Environmental growth parameters such as temperature and aeration must be regulated.

The proper use of an organism to form distinct products of defined quality, and in large amounts, needs a detailed knowledge of the biochemical mechanisms of product formation.

In its simplest form, the bioprocess can be seen as just mixing microorganism with a nutrient broth and allowing the components to react eg yeast cells with a sugar solution to produce alcohol. More advanced and sophisticated process operating at large scale need to control the entire system so that the bioprocess can proceed efficiently and be readily and exactly repeated with the same amounts of raw materials and innoculum to produce precisely the same amount of product.

Bioreactor/Fermenter

All biotechnological processes are essentially performed within containment system or Bioreactor. For each fermentation process, the most suitable fermenter (or bioreactor)

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must be designed to give correct environment for optimization of organism growth and metabolic activity.

Bioreactors range from simple stirred or non-stirred open containers to complex, aseptic, integrated systems involving computer inputs. There is the need to avoid contaminant of the fermentation process by unwanted microorganisms and also avoid undue release of the culture organism into the environment.

There are 2 distinct types of bioreactors, viz:

1. Non-Aseptic Systems, where it is not necessary to operate with entirely pure culture eg in brewing.
2. Aseptic Systems, where pure cultures are essential for product formation eg antibiotic, vitamins and polysaccharides formation.

Standards of Materials in Fermenter Design

The standards of materials used in sophisticated fermenter design are:

1. All materials coming in contact with the solutions entering the bioreactor or the actual organism culture must be corrosion resistant to prevent toxic metal contamination of the process.
2. The materials must be non-toxic so that slight dissolution of the material or component does not inhibit culture growth.
3. The materials must withstand repeated sterilization with pressure system.
4. The bioreactor stirrer system, entry ports and end plates must be easily machinable and sufficiently rigid not to be deformed or broken under mechanical stress.
5. Visual inspection of the medium and culture is advantageous and hence transparent materials should be used wherever possible.

Factors Affecting Microbial Culture Growth in a Bioreactor

1. Nutrient Composition

Microorganisms require nutrients [water, source of energy (sugars, alcohols and fats), source of nitrogen (e.g. amino acids), vitamins and minerals] for growth. The nutrient composition in the culture depends on the product to be produced and the microorganism in it.

Carbohydrate fermenting microorganisms are usually lactic/acetic acid bacteria, yeast and moulds. Proteins are required by proteolytic organisms especially *Bacillus*,

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Pseudomonas and *Proteus*. Fats are digested by relatively few microorganisms, mainly moulds and few Gram- negative bacteria.

Bacteria and molds are able to synthesize their vitamin B and hence able to grow in foods with low vitamin B e.g. in fruits

2. pH/Acidity

Most microorganisms grow best at pH values around 7.0 (6.6 – 7.5). The pH of acid food is below pH 4.5, and that of non-acid food above pH 4.5. Most fruits are acid foods, while nearly all vegetables, fish, meats are non-acid foods.

Acid foods e.g. fruits are mostly utilized by yeast and fungi because they can grow at pH values < 3.5, while non – acid foods e.g. vegetables and meat are utilized by bacteria.

3. Presence and Concentration of Oxygen/Air

Moulds and aerobic bacteria grow only where there is plenty of air. Yeast and facultative bacteria such as *Leuconostoc* can grow in closed containers as well as when exposed to the air. *Clostridia* and other obligate anaerobes grow under strict anaerobic condition. For the facultative aerobes, suitable concentration of oxygen is required in the bioreactor.

4. Temperature

The lowest temperature at which a microorganism has been able to grow is – 34°C; the highest been 90°C. Psychrotrophs (eg *Bacillus*) can survive low temperatures (< 7 - 20 °C), mesophiles (eg *Lactobacillus*) require warm temperatures (20 – 45°C) and thermophiles (eg *Streptococcus thermophilus*) require high temperature (45 – 65°C). Each fermenting organism has an optimum temperature for growth and product formation.

Operating Guidelines for Bioreactor System

The operating guidelines for the optimization of the bioreactor system are:

1. The bioreactor should be designed to exclude entrance of contaminating organisms as well as containing the desired organisms.
2. The culture volume should remain constant i.e. no leakage or evaporation.
3. The dissolved oxygen level must be maintained above critical levels of aeration and culture agitation for aerobic organisms.

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4. Environmental parameters such as temperature, pH etc must be controlled and the culture volume must be well-mixed.

GENETIC IMPROVEMENTS OF FERMENTATION PRODUCTS

The most widely consumed products of fermentation process are beers and wines. The starting materials in brewing are mostly cereals like barley, wheat, millet etc, which are poor sources of proteins. In addition, the biological value of their proteins is low as a result of deficiency of some essential amino acids. Fermentation process is now genetically improved in order to enhance the protein contents of the products through:

1. Genetic Modification of Fermenting Organisms

The increase in the quantity of essential amino acids in the fermentation products of cereal grains by *in vivo* genetic manipulation of the fermenting organisms, such as *Sacchromyces* spp and lactic acid bacteria with mutants of the organisms producing larger amounts of the deficient essential amino acids, like lysine in sorghum products, and lysine, tryptophan and methionine in maize products.

2. Transforming Cereals into Nitrogen-Fixers

Gene transplantation techniques, such as protoplast fusion, are now offering the opportunity for cereals to be transformed into nitrogen – fixers. Legumes are plants with richest amounts of proteins because they harbor nitrogen –fixing bacteria in their root nodules. These bacteria, like *Azotobacter*, have nitrogenase enzyme, which can convert nitrogen to NH_3.

$$N_2 + 10H^+ + 8e^- + 16ATP \rightarrow 2NH_4^+ + 16ADP + 16\ Pi + H_2$$

The NH_4^+ produced is used in the synthesis of amino acids in the bacteria, which are also taken up by the plants.

The gene coding for the nitrogenase enzyme is now being planted in cereals through recombinant DNA technology so that they are conferred with the ability to fix nitrogen. This improves the levels of amino acids and proteins in the cereals and hence in their fermentation products.

ENZYME TECHNOLOGY

This is a type of biotechnology in which enzymes are used in industrial production of materials. It is now contributing hugely in providing solutions to societal challenges eg in

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ood production, energy shortage and preservation and improvement of the environment.

Enzyme technology embraces production, isolation, purification, use in soluble form and finally the immobilization and use of enzymes in a wide range of bioreactor system.

The challenges in isolation and purification of enzymes involve choice of source, method of homogenization and methods of separation.

Some of the applications of enzymes in the industry include:

Industry	Enzymes used and Reaction catalyzed	Uses
Baking	Fungal α-amylase (endo hydrolysis of α-1,4 glycosidic linkage in starch)	Breakdown of starch in flour to sugar, used by yeast in the production of white bread, burns, rolls etc.
	Proteases (hydrolysis of peptide bonds in proteins).	Used in biscuit manufacture in order to lower protein level of flour.
Brewing	Enzymes produced from barley during mashing stage of beer production	Degrade starch and proteins to produce simple sugars, amino acids and peptides used by yeast to enhance alcohol production.
	-Industrially Produced Enzymes -Amylases, glucanases, proteases	Split polysaccharides and proteins into maltose and peptides.
	-β- glucanase (breaks walls of endosperm and seeds).	-Improve filtration characteristics.
	-Amyloglucosidase (exo-hydrolysis of α-1,4 glycosidic bonds in starch)	- In production of low calorie beer.
	-Proteases	- Remove cloudiness during beer storage.
Dairy		

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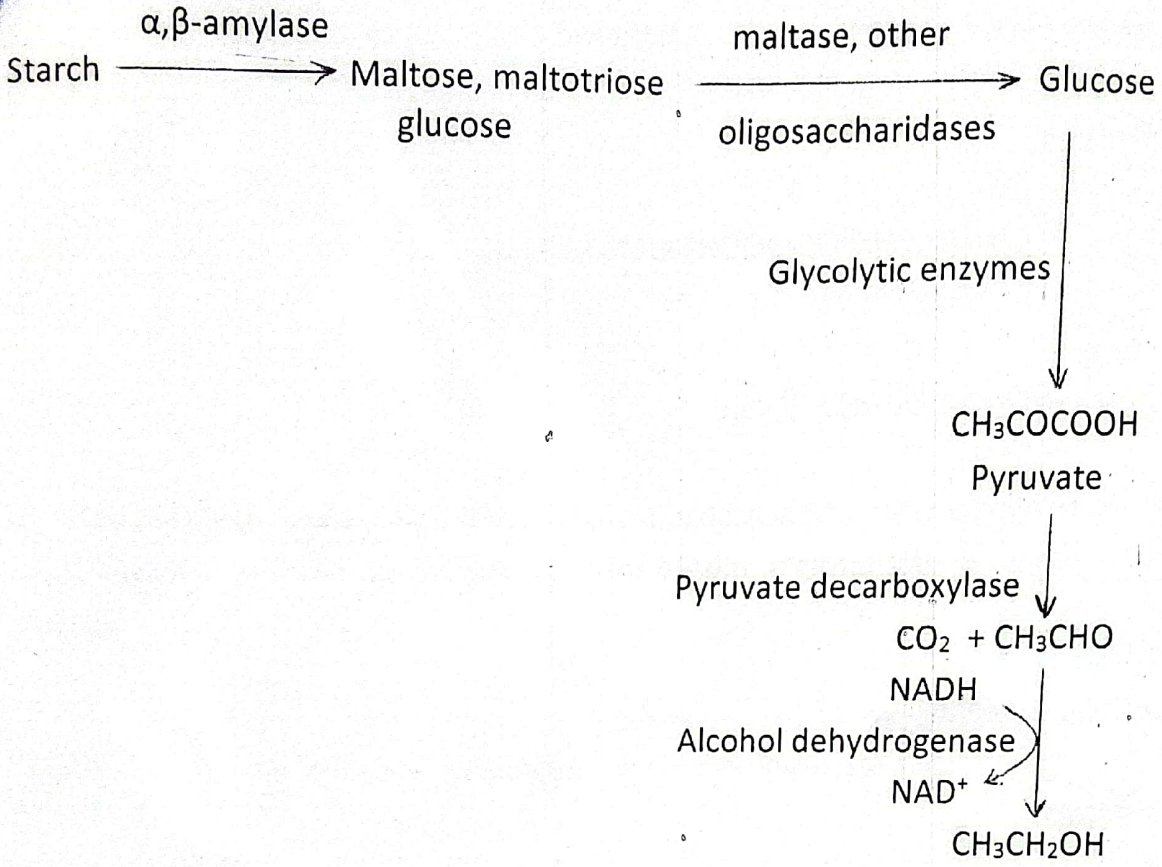
	-Rennin (derived from stomach of young ruminants - calves, lambs, kids; hydrolyses peptide bond in casein).	Used to split casein, the protein in milk, in the production of cheese.
	-Microbiologically Produced Enzymes	
	-Fungal proteases	-Used as substitute for rennin in cheese production.
	-Lipases (hydrolysis of triglycerides to fatty acids and glycerol).	-Enhance ripening of blue mould cheese.
	-Lactases (hydrolyses lactose to glucose and galactose).	-In ice-cream production and lactose-free milk.
Starch	-Amylases, amyloglucosidases etc	-Convert starch into glucose and various syrups.
	-Glucose isomerase (isomerizes glucose to fructose).	- High-fructose syrup production
Meat	Proteases especially papain, trypsin and bromelain	-Used for tenderization and marination of meat before packing

USE OF MICROORGANISMS AS SOURCES OF ENZYMES

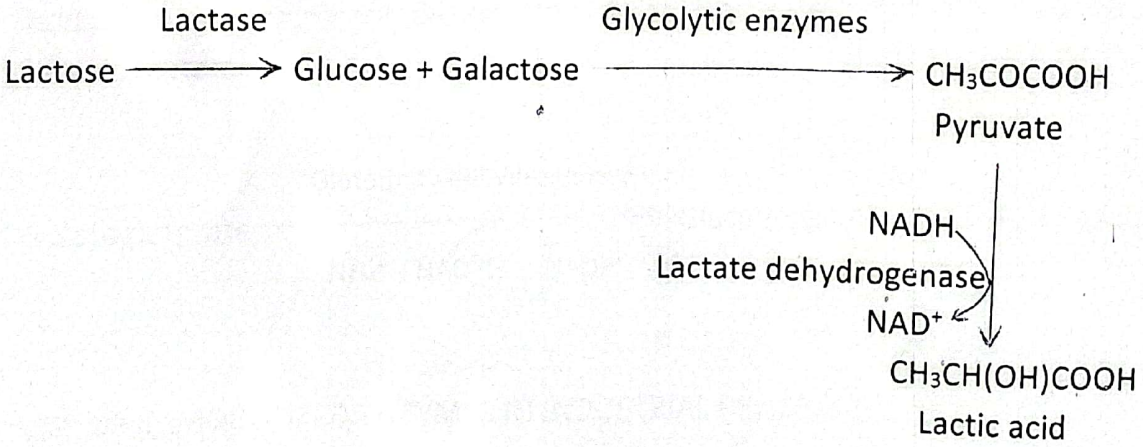
In many industrial processes, pure enzymes are not used to produce the desired products but rather microorganisms are used as sources of the enzymes.

a. **Brewing**: In brewing, the organisms used are barley and *Saccromyces* species.
When allowed to germinate, barley contains endo-1,3-β-D-glucanase, which breaks walls of endosperm or seeds; and amylases, which hydrolyze starch to maltose, maltotriose and glucose.
Saccromyces species contains oligosaccharidases (which convert maltose, maltotriose to glucose), all glycolytic enzymes (which convert glucose to pyruvate), pyruvate decarboxylase (which converts pyruvate to acetaldehyde) and alcohol dehydrogenase (which converts acetaldehyde to ethanol).
In wine – making, the starting material is mainly mono- or disaccharide.

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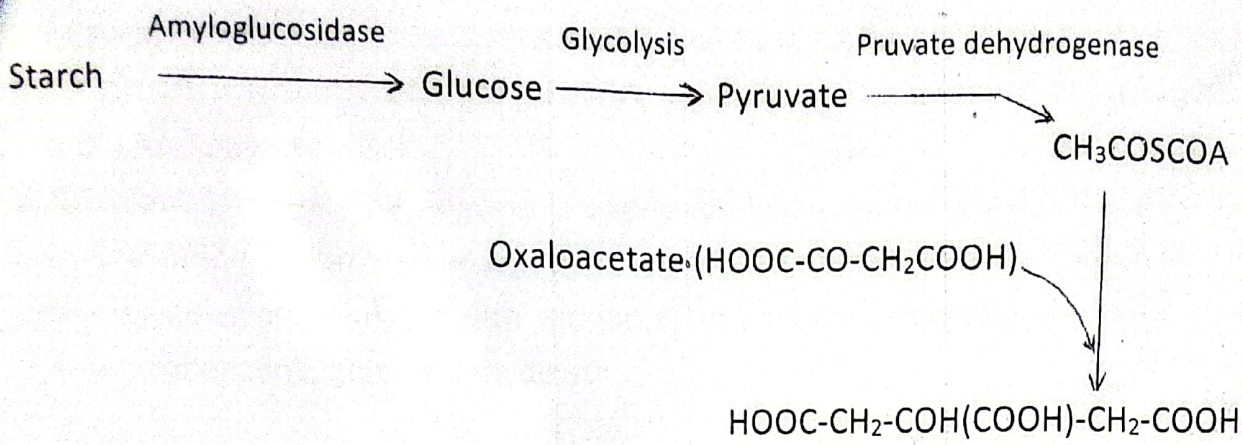


b. **Cheese – Making**: The organisms used are *Streptococcus lactis* or *Streptococcus cremoris*, which have enzymes that can convert lactose to lactic acid.

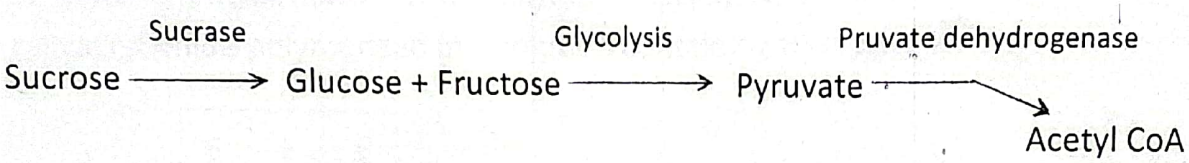


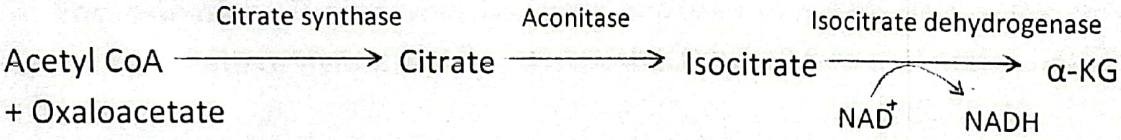
c. **Citric Acid Production**: The organism used is *Aspergillus niger*, which has all enzymes that convert starch to citric acid. The citric acid is used in soft drinks, jams, jellies, flavouring and leather tanning. It lowers the pH of foods to below 4.5 in order to achieve sterilization and inhibition of growth of pathogenic microorganisms like *Clostridium botulinum*.

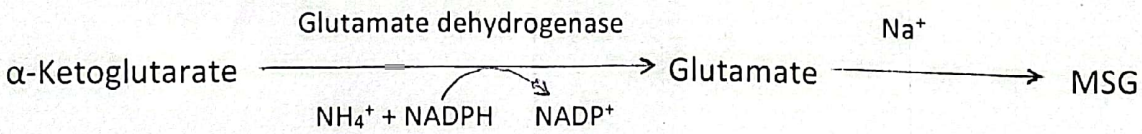
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d. **Monosodium Glutamate**: The organism used *Aspergillus* species has all the enzymes that convert sucrose to glutamate. Monosodium glutamate is used in seasoning of foods.







USE OF IMMOBOLIZED ENZYMES IN INDUSTRIAL PROCESSES

In order to reduce the cost of using enzymes in industrial processes, the enzymes are now being immobilized. Immobilized enzymes are enzymes attached to an insoluble support in order to allow their reuse and continuous use without difficult recovery process.

Methods of Immobilization

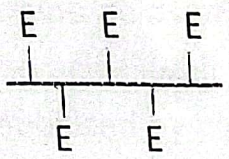
The ways in which enzymes can be immobilized are:

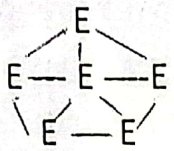
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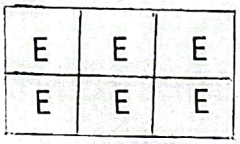
1. **Adsorption** in which the enzyme is adsorbed by weak forces, such as electrostatic and hydrophobic interactions on the insoluble support such as alumina, bentonite, cellulose and ion-exchange resins.

2. **Covalent Linkage:** The enzyme is covalently linked to the insoluble support such as polyacrylamide, nylon, cellulose, sephadex etc. The enzyme can be linked together by intermolecular crosslinkage with the aid of bifunctional reagents such as 1,5-diflouro-2,4-dinitrobenzene, glutaraldehyde etc .

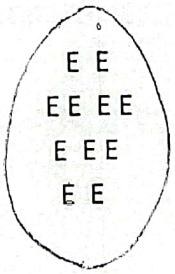




3. **Matrix Entrapment**: The enzyme is trapped in pores of a matrix, such as polyacrylamide polymerized in a solution containing the enzyme.



4. **Encapsulation**: The enzyme becomes enclosed in a capsule usually generated from cellulose nitrate or nylon, which is permeable to small ions and molecules.



Other methods include

- Affinity binding
- Cloning (cloning the genes of enzyme alongside that of matrix)
- Metal-linked method (cationic metal binding to negatively charged enzyme)

Application of Immobilized Enzymes

1. Production of High-Fructose Syrup from Corn Starch

High-fructose syrupis being produced on a large scale, and is gradually displacing sucrose in many applications because fructose is sweeter and does not crystallize easily.

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$$\text{Starch} \xrightarrow{\text{Amyloglucosidase}} \text{Glucose} \underset{}{\overset{\text{Glucose isomerase}}{\rightleftharpoons}} \text{Fructose}$$

Amyloglucosidase is obtained from *Aspergillus niger*, and is being immobilized on porous silica. Glucose isomerase is obtained from *Streptomyces* species, and is being immobilized by attachment of DEAE cellulose or to a porous ceramic carrier.

2. **Production of L-Amino Acids from Racemic Mixtures**

DL-amino acids are normally produced by chemical synthesis. They are converted to L-amino acids (which are preferred by biological systems) by aminoacylase. The enzyme is immobilized by binding to DEAE-sepadex columns.

3. **Chymosin in Cheese-Making**

Chymosin is obtained from calf stomach, and is being immobilized by entrapment with resin pre-polymers or attachment to DEAE sephadex..

4. **Pectinase in Extraction and Clarification of Fruit Juice**

The enzyme is obtained from *Mucor, Penicillum*, and *Aspergillus* species. It catalyses the random hydrolysis of α-1,4 galactosiduronic linkages in pectin. It is used in the extraction and clarification of wines and fruit juices. It is immobilized on calcium alginate and on co – precipitated magnetic nanoparticles.

Other applications include:

- Biodiesel Production (using lipase to produce methylester from vegetable oil in the presence of methanol).
- Bioremediation
- Protease in Detergent
- Biosensor Technology

USE OF GENETIC ENGINEERING IN ENZYME TECHNOLOGY

Recombinant DNA Technology is applied in Enzyme Technology in the following ways:

1. **Transfer of Enzyme Gene from One Organism to Another**

The gene coding for an enzyme identified as suitable for industrial use can be cloned into a more suitable production host organism. Industrial fermentation can then be carried out. For example, the enzyme lipolase was identified in the fungus *Hamicola*

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languinosa at low levels. The gene for the enzyme was cloned into the production fungus *Aspergillus oryzae* and commercial level of the enzyme was produced.

2. **Protein Engineering**

This is the use of recombinant DNA technique to alter the catalytic and other properties of enzymes. Some of the objectives of modification of enzymes include:

- Enhancing the activity of the enzyme
- Improving the stability of the enzyme
- Permitting the enzyme to function in a changed environment
- In order to change the pH an temperature optima
- Changing the specificity of an enzyme so that it catalyses the conversion of different substrate
- Changing the reaction catalysed

ADDITIVES USED IN INDUSTRIAL PROCESSING OF FOODS

Food additives are substances incorporated into processed foods for functional purposes, such as improved keeping quality, enhanced nutritional value, functional property provision and improvement, processing facilitation etc.

Some of the commonly used food additives include:

1. ACIDS

Organic and inorganic acids are used as food additives for numerous purposes such as buffering systems, chemical leavening, microbial inhibitors in food preservation (eg sorbic and benzoic acids), as chelating agents etc. Examples are:

a. **Lactic Acid** ($CH_3CH(OH)COOH$) causes coagulation of casein (milk protein) in the production of cheese.

b. **Gluconic Acid and its Lactone** (δ-Gluconolactone) is used for slow acid production in cultured dairy products.

c. **Citric Acid** is added to moderately acidic fruits and vegetables in order to achieve sterilization and inhibition of growth of hazardous microorganisms.

d. **Tartaric Acid** (HOOC-CHOH-CHOH-COOH) and potassium acid tartrate are employed in the production of fondant and fudge to induce limited hydrolysis of sucrose.

e. **Phosphoric Acid** (H_3PO_4) is the most widely used food acidulant employed in flavored carbonated beverages in colas and root beer.

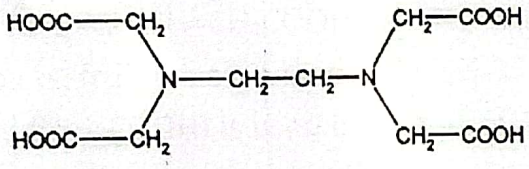
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2. CHEMICAL LEAVENING AGENTS

These are chemicals that can react to release gas in a dough or batter under appropriate conditions of moisture and temperature. CO_2 is the gas generated from currently used leavening systems. The most common leavening salt is sodium bicarbonate ($NaHCO_3$).Others are ammonium carbonate [$(NH_4)_2CO_3$], sodium aluminum sulfate [$Na_2SO_4.Al_2(SO_4)_3$], dicalcium phosphate dehydrate [$Ca_2HPO_4.2H_2O$] etc.

3. CHELATING AGENTS (SEQUESTRANTS)

Chelating agents are able to form complexes with metals and alter their properties and effects in foods, thereby aiding in food stabilization. They include natural substances such as polycarboxylic acids (citric, malic, oxalic and succinic acids), polyphosphoric acids (ATP and pyrophosphate) etc. Chelating agents used in food include salts of ethylenediaminetetraacetic acid (EDTA).

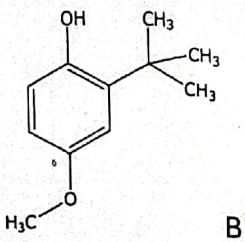


4. ANTIOXIDANTS

These are substances which prevent oxidation of biomolecules. Their use as food additives is to prevent oxidation reactions in foods. Some antioxidants added to foods include:

Natural: - Tocopherols
- Gossypol (occurs naturally in cottonseed)
- Coniferyl alcohol (found in plants)
- Guiaconic and Guiacic Acid (from gum guaiac)

Artificial: – Butylated Hydroxyanisole
- Butylated Hydroxytoluene
- Tertiarybutylhydroquinone
- Propylgallate



Butylated hydroxyanisole

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5. ANTIMICROBIAL AGENTS

These are chemical preservatives with antimicrobial properties added to foods for prevention of spoilage and safety assurance. They include:

a. Sulfites and SO_2

   SO_2 and its derivatives, sulfite (SO_3^{2-}), bisulfate (HSO_3^-) and metabisulfite ($S_2O_5^{2-}$) are added in foods to prevent browning, inhibit and control microorganisms and as antioxidants.

b. Nitrite and Nitrate in meats

c. Acetic Acid (CH_3COOH) in form of vinegar (4% acetic acid) and its salts (sodium acetate, potassium acetate etc) in bread, baked foods, pickled meats and fish products.

d. Propionic Acid (CH_3CH_2COOH) and its sodium and calcium salts exert antimicrobial activity against moulds and some bacteria - used in Swiss cheese.

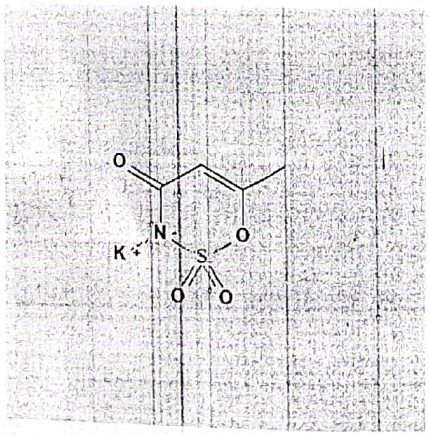
e. Sorbic Acid (CH_2=CH-CH=CH-COOH) is used against moulds and yeasts in cheese, baked products, fruit juices, wines and pickles.

f. Benzoic Acid (C_6H_5COOH) is used in fruit juices, carbonated beverages, pickles and sauerkraut.

6. LOW – CALORIE SWEETENERS

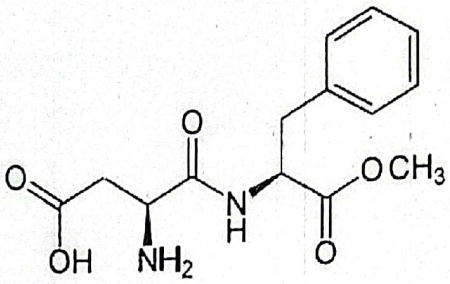
These are intensely sweet non-nutritive sweeteners added to some foods. Some examples are:

a. Acesulfame K (Sweetness Value, SV, = 200)

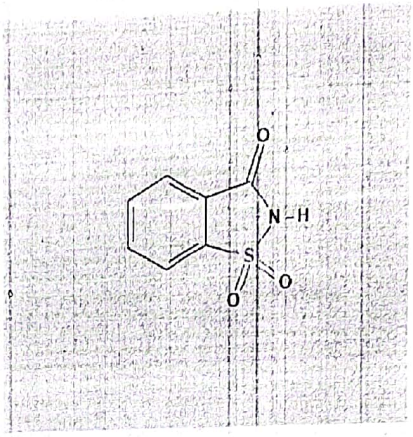


b. Aspartame (SV= 180-200)

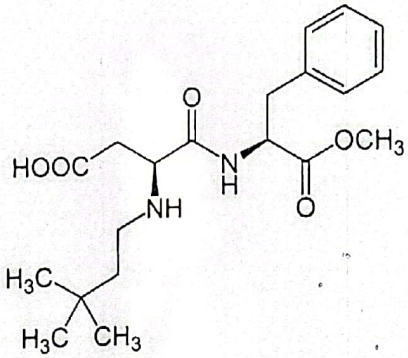
CamScanner



c. Saccharin (SV= 300-400)

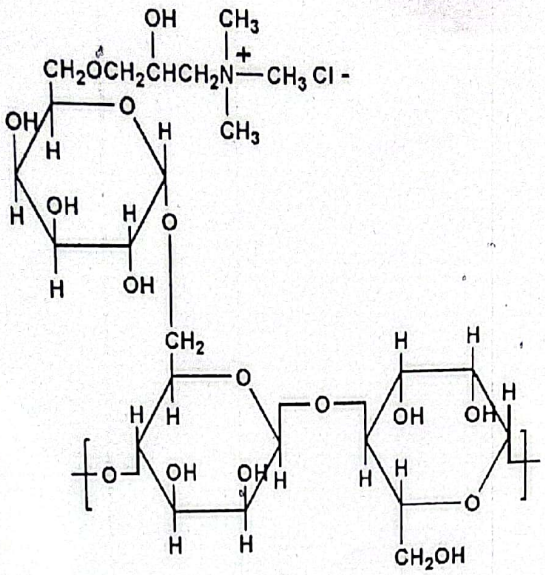


d. Neotane (SV= 7,000 – 13,000



7. STABILIZERS AND THICKENERS

These are hydrocolloid materials used in foods to provide stabilizations for emulsions, suspensions and foams for general thickening properties. They are mostly polysaccharides such as gum Arabic, guar gum, carboxymethylcellulose, carrageenan, agar, starch and pectin.



Repeating unit of Guar Gum